

Aryan Sharma

SENIOR AI/ML ENGINEER · IIT Kanpur

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SUMMARY

Senior ML Engineer with nearly 5 years of experience at Jaguar Land Rover designing, training, and deploying production-grade deep learning models, distributed infrastructure, and automated data pipelines Expert in 3D computer vision, MLOps orchestration, and cost-efficient cloud architectures.

WORK EXPERIENCE

JAGUAR LAND ROVER

Bengaluru, India

Senior AI/ML Engineer (Graduate SWE → AI/ML Engineer → Senior AI/ML Engineer)

Jul 2021 – Present

Scene Search & 3D Perception

- Built **Scene Search**, a semantic data-curation platform from scratch; reduced target-scenario discovery time from ~50 hours to 1.25 hours (**97.5% reduction**) using CLIP/VLM embeddings and Qdrant/Faiss over a Voxel51 UI on AWS EKS
- Engineered an in-house **monocular 3D object-detection** network; increased 2D-to-3D localization accuracy from **55% to 91%** and deployed it as the core API driving the internal Ground Truth System.
- Developed Ray Serving enabled auto-labelling pipeline for **LiDAR point-cloud panoptic segmentation** benchmarked on nuScenes datasets, leveraging Segment Anything (SAM) one-shot masks, kNN spatial clustering for Ground-Truth Data
- Implemented a Redundancy-Aware Frame Selection strategy for video-embedding generation, dramatically optimizing indexing and storage efficiency for long driving sequences

ML Platform & MLOps Engineering

- Templated enterprise ML infrastructure using **Infrastructure as Code (AWS CDK)** and orchestrated complex data validation and batch processing using **Argo Workflows/Events**
- Designed a reusable **Ray distributed-training** framework with on-demand EKS RayClusters, integrating ClearML logging and automated node teardown to eliminate idle GPU compute costs
- Established a centralized **CI/CD** automation framework across **24+ core repositories** and deployed a unified GPU/CPU observability stack (OpenTelemetry + Grafana) spanning on-prem DGX systems and AWS cloud instances
- Created a GitLab-triggered automated training-to-deployment pipeline using AWS Step Functions, SageMaker, Lambda, and S3; optimized EKS compute nodes using Fargate and EC2 auto-scaling-to-zero

AGROSTAR

Pune, India

Machine Learning Intern

May 2019 – Jul 2019

- Developed an unsupervised **speaker-diarization** engine (K-Means clustering) processing over 10k daily Indic language call-center logs, reducing operations costs by **76.2%** and turn-around time by **50%**

SKILLS

ML / Deep Learning	LLMs, VLMs, RAG, Transformers, Vector Search, Computer Vision, 3D Object Detection, Panoptic Segmentation, Distributed Training
MLOps & Infrastructure	Infrastructure as Code (IaC), AWS CDK, Kubernetes (EKS), Ray, Argo Workflows/Events, Docker, CI/CD, OpenTelemetry, Grafana, ClearML
Cloud Platforms	AWS (EC2, Fargate, Lambda, Step Functions, SageMaker, S3, EFS, FSx, CloudWatch)
Data Systems	Qdrant, Faiss, MongoDB, Voxel51 / FiftyOne
Languages & Tools	Python, C++, Bash/Shell, MATLAB

EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY, KANPUR

Kanpur, India

B.Tech – M.Tech (Dual Degree) in Aerospace Engineering | CPI: 7.7/10

Jul 2016 – May 2021

- Publication:** A. Sharma, N. Jain, M. Kothari, pre-print: “Lightweight Multi-Drone Detection and 3D-Localization via YOLO”
- Relevant Coursework:** Machine Learning, Linear Algebra, Probability & Statistics, Data Structures & Algorithms

SELECTED PROJECTS/OPEN SOURCE

- Synthetic Drone Dataset generation pipeline:** Released multi-drone synthetic dataset along with data generation pipeline based on **AirSim** simulation Github Repo: github.com/aryanshar/swarm-detection.

AWARDS & ACHIEVEMENTS

- Global Technical Specialist Conference, JLR (2025):** Selected internationally to present a research poster and PoC on agentic MLOps onsite at JLR UK
- Creators Kudos & Exceptional Creator Award, JLR (2024):** Received for outstanding contributions and technical innovation across various core enterprise infrastructure projects
- Global Winner, JLR Hackathon (2023):** Awarded 1st place worldwide; engineering solution was judged best architecture for 360-degree object detection using a camera-only sensor suite
- JEE Main & Advanced (2016):** Secured **AIR 1936** (Main) and **AIR 2925** (Advanced) out of 1.5 million candidates